Evolutionary Algorithm to *ab initio* Protein Structure Prediction with Hydrophobic Interactions

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***Abstract*— Proteins are polymers whose chains are composed of 20 different monomers, called amino acids. The problem of Protein Structure Prediction (PSP) is the determination of protein 3D conformation from its amino acid sequence. Two main strategies are usually employed to work with PSP: homology and *Ab initio* approaches. This paper presents an Evolutionary Algorithm to PSP using an *Ab initio* approach (ProtPred). The predictions are evaluated using fitness functions based on potential energies (electrostatic and van der Waals) and hydrophobic interactions. The proposed approach uses dihedral angles and main angles of the lateral chains to model a protein structure. ProtPred is evaluated using relatively complex cases for an *Ab initio* approach. Results have shown that ProtPred is a consistent approach.**

I. Introduction

Currently several proteins are used as drugs. Prominent examples are the known protein insulin, the major drug against diabetes or, more recently, the anti-HIV protein T20 [1]. However, sometimes in order to use proteins as drugs is necessary to know their tertiary structure (3D conformation), that determines their function. In the last years, several amino acids sequences of proteins have been obtained, however their tertiary structure remain unknown.

PSP is a computationally open problem and several methodologies have been investigated to solve it. The interest in discovering a methodology to solve PSP extends into many fields of research, including biochemistry, medicine, biology, engineering and scientific disciplines. Native protein structures have been determined using X-ray crystallography methods and magnetic nuclear resonance [2]. The latter has its application restricted to proteins with small size, while the former needs a great amount of laboratory processing that requires high cost.

On the other hand, approaches for PSP range from empirical researches to mathematical modeling for protein potential energy. An algorithm strategy to solve PSP uses information from protein homology to guide a search process. Despite the relevant results obtained using such strategies, algorithms based on protein homology are highly dependent on the set of proteins with native known structure. This set is extremely smaller than the universe of proteins [3].

On the other hand, *Ab initio* PSP does not depend on previous knowledge of protein structures. This is one of the most important unsolved problems in molecular biophysics [4]. At first glance, it may not seem complex since by knowing the exact formulation of the physical environment within a cell, where proteins fold, it is possible to mimic the folding process in nature by computing the molecular dynamics based on our knowledge of the physical laws [5]. Nevertheless, we do not completely understand the driving forces involved in protein folding. Perturbations in the potential energy landscape may result in a different folding pathway, generating a different 3D structure.

Due to insufficient results obtained for the tertiary protein structure from amino acid sequences, many different computational algorithms have been investigated. Among these algorithms, Evolutionary Algorithms (EAs) have presented relevant results [6] [7] [8] [9] [10] [11] [12]. EAs are powerful tools of optimization inspired in natural evolution and have been applied to many complex problems in the most different areas of human knowledge [13].

Another aspect of PSP is that it is a multi criterion problem since several energies should considered in order to evaluate biochemistry interactions in a protein structure [3]. A complete model of forces acting is hard to computationally implement. For example, the protein interaction with the solvent is very complex. It depends on how solvent components change their distribution in relation to the protein molecule during the folding. Due to difficult like that, approximated models are used. This implies that only main energies are employed.

This paper presents an evolutionary *Ab initio* approach to PSP (called ProtPred) using as fitness potential energies and the hydrophobic interaction between the amino acids. Section II presents basics concepts of evolutionary algorithms. Section III introduces the PSP problem and its characteristics. In Section IV is presented ProtPred. The results obtained can be seem in Section V and Section VI presents the conclusions of this work.

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II. Evolutionary Algorithms

EAs are a broad class of optimization algorithms that attempt to simulate the inner workings of evolution. Evolutionary approaches are based on the collective learning process of a population of individuals. Each individual represents a search point in the space of potential solutions to a given problem [13].

Descendants of individuals are generated by stochastic process, which in general intended to model mutation and recombination (or crossover). Mutation can be viewed as an erroneous self-replication of individuals, while recombination exchanges information to or more existing individuals.

By means of evaluated individuals in their environment, a measure of quality or fitness value can be assigned to individuals. According to the fitness measure, the selection process favors better individuals to reproduce more often than those that are relatively worse [13].

III. Protein Tertiary Structure Prediction

Proteins are macro molecules built from 20 basic units, named amino acids (or peptides). All amino acids are similar since they possess a same basic generic chemical structure. There is a central carbon atom (Cα) angle attached to a hydrogen atom, an amino group (NH_2), a carboxyl group (COOH) and a lateral chain or residue (R), which distinguishes one amino acid from the others. Every residue is assigned either a 3-letter or a 1-letter code. During DNA transcription-translation phases, the carboxyl group of one amino acid is joined with the amino group of another, forming a peptide bond, to release water. Thus, we can refer to a protein as a polypeptide formed by a backbone or a main chain (the sequence of peptide bonds) and a side chain (the sequence of residues) [14].

The structure of a protein is hierarchically represented by three structural description levels. The primary structure is the sequence of amino acids in the polypeptide chain, which can be described as a string from a finite alphabet. The secondary structure refers to the set of local conformation motifs of the protein and schematic the path followed by the backbone in the space. The most important description level and main objective of experimental and prediction efforts is to obtain the protein's tertiary structure. It describes the three-dimensional organization of polypeptide chain atoms.

The formation process of the tertiary structure is designated *folding*. There are some physical properties that define this process [15]:

1) Hardness of the backbone of the sequence;
2) Interactions between amino acids, including electrostatic interactions;
3) van der Waals forces;
4) Volume constraints;
5) Hydrogen or disulfate bounds;
6) Interactions of amino acids with water (hydrophobic effect).

The interactions of amino acids with water is very important for stabilizing a protein conformation. They are called hydrophobic interaction [16]. Hydrophobic interaction refers to the physical property of a molecule that is repelled from a mass of water. Hence, the hydrophobic amino acids of a protein will tend to cluster together, not as a result of attraction, but as a result of their repulsion by the hydrogen bonded water network in which the protein is dissolved [17].

Due to the difficulty in understanding the folding process of a protein, this problem has been modeled in the literature as an optimization process. Different computational strategies have been investigated to solve this problem: homology [2] and *Ab initio* [18]. This paper will focus on the *Ab initio* approach.

The *Ab initio*[1] structure prediction aims to predict a protein structure only from its amino acid sequence. A protein sequence is generally assumed to fold to a native conformation or an ensemble of conformations that is near the global free-energy-minimum [18].

In an *Ab initio* approach, no homology between proteins is employed. As consequence, *Ab initio* prediction is more challenging than homology modeling. This paper focus on *pure Ab initio* approach, i. e., neither homology nor prior knowledge about any relation between sequences and 3D conformation are employed to guide the search algorithm[2]. Moreover, *Ab initio* prediction is the only way to derive a prediction, when no similar test fold is known. On the ,ther hand, pure *Ab initio* prediction could avoid prediction biased by any prior knowledge since, nowadays, it is relatively restricted.

The dihedral angles ϕ and ψ of each residue determine the protein fold. Unfortunately, there is a large number of possible value for these angles, making the *Ab initio* approach computationally very complex.

IV. Proposed Approach

In works with PSP, two important tasks must be defined: a good representation of the conformations and cost function to evaluate conformations. These aspects are discussed in the next subsections.

A. Representation of the conformation

There are few conformations and representations commonly used [3]:

- all-atom three-dimensional coordinates;
- all-heavy-atom coordinates;
- backbone atom coordinates and side-chain centroid;
- C_α coordinates;
- backbone and side-chain torsion angles;
- lattice models.

In our approach, we chose the backbone and side-chain torsion angles representation since it produces the most precise model. The covalent bond lengths and angles are considered at their ideal values [3]. The dihedral angle ω is

[1] From Latin: "from beginning".

[2] Even when no homology proteins are found, homology between their subsequences can be employed to derive predictions of small secondary structures, witch can be used to catalize PSPs [19].

fixed at 180^o. Thus, in order to represent a solution (protein prediction) we need two torsion angles of the backbone (ϕ and ψ) and the side-chain torsion angles (χ_i $i = 0, \ldots, 4$, depending on each residue [3]) for each residue in the protein. An array with ϕ, ψ and χ_i values for each residue is a chromosome.

B. Cost Function

To evaluate the protein's fold, it is necessary to use a cost or energy function. Quantum mechanic produces the most adequate energy functions, but they are too computationally complex to be employed in modeling larger systems. Thus, the proposed approach uses energy function obtained from classical physics, called potential energy functions or force fields and a function that consider the hydrophobic interaction. These functions provided an energy value based on the molecule conformation.

1) Potential Energy Function: The most typical potential energy functions have the following form [20]:

$$Energy(R) = \sum_{bonds} B(R) + \sum_{angles} A(R) + \sum_{torsions} T(R) + \sum_{non-bonded} N(R) \quad (1)$$

where R is a vector representing the molecule conformation, typically in Cartesian coordinates or torsion angles.

The literature on cost functions is enormous [20] [21]. ProtPred proposed to PSP problem is based on TINKER (Software Tools for Molecular Design) energy functions. These functions requires a series of parameters. The best source of energy function parameters found in the literature are available in the CHARMM (Chemistry at HARvard Macromolecular Mechanics) parameters $v.27$, which was used by ProtPred. TINKER package is a composite sum of several molecular mechanical functions. Commonly approaches to PSP problem use sum of all potential energy functions available in TINKER software [8]. However, previously experiments show that only two potential forces, van der Waals and electrostatic, can give an adequate model to evaluate individuals. For this reason, ProtPred uses only van der Waals and electrostatic energies as following (see potential energy functions in Equation 4):

$$\begin{aligned} E &= \sum_{non-bond} \varepsilon_{i,j} \left[\left(\frac{R_i + R_j}{r_{i,j}} \right)^{12} - 2 * \left(\frac{R_i + R_j}{r_{i,j}} \right)^{6} \right] \\ &+ \sum_{non-bond} \frac{q_i q_j}{D r_{i,j}} \end{aligned} \quad (2)$$

where

- (i) $\varepsilon_{i,j}$ is the Leonard-Jones well depth, $r_{i,j}$ is the distance between atoms i and j, R_i is the van der Waals atom i radius, R_j is the van der Waals atom j radius;
- (ii) q_i and q_j are the partial atomic charges from atom i and j and D is the dielectric constant.

2) Hydrophobic Interaction Energy Function: The hydrophobic interaction function is based on the HP Model, where the alphabet size of protein can be reduced to a lesser number classifying the amino acids in two types: H and P. The energy function for this system favors interactions between HH monomer types and is indifferent to all the others (HP and PP). It aims to highlight the importance of the hydrophobic effect in protein folding. In an aqueous medium, globular proteins tend to have a core of hydrophobic amino acids, surround by polar residues on the surface. In the HP model, the H monomers correspond to hydrophobic residues that collapse to form a core surrounded by polar, P, monomers.

The energy function for this model is:

$$\begin{aligned} E(\{r_1 \ldots r_N\}) &= -|\epsilon| \sum_{i,j=1;i<j}^{N} \Delta(r_i - r_j) \\ &+ \epsilon_2 \sum_{i,j=1;i<(j-1)}^{N} \delta(r_i - r_j) \end{aligned} \quad (3)$$

$$\Delta(r_i - r_j) = \begin{cases} 1 & \text{if } i, j \text{ both H-type and nearest neighbour;} \\ 0 & \text{otherwise} \end{cases}$$

$$\delta(r_i - r_j) = \begin{cases} 1 & \text{if monomers } i \text{, occupy same site;} \\ 0 & \text{otherwise} \end{cases}$$

where:
$|\epsilon| \equiv$ strength of HH attraction.
$\epsilon_2 \equiv$ energetic penalty parameter for sites containing two or more monomers.

This hydrophobic energy function is very used in lattice representation of the conformation [22]. ProtPred uses the dihedral angle representation, thus, in order to determine if a residue is near of other, it is defined a maximum distance between their C_α's.

Equation 4 represent our minimization objective, composed by the two potential energy functions (van der Waals and electrostatic) and the hydrophobic energy that simulate the interaction between protein and environment. The torsion angles of a protein are the decision variables of the PSP problem. The constraint determines the feasible range for each variable.

$$Energy = E_{vdw} + E_{elec} + E_{hydrophobic} \quad (4)$$

C. ProtPred

The proposed ProtPred is based on a conventional EA. The algorithm starts initializing a random conformation. The torsion angles (ϕ, ψ, χ_i) are generated at random from constrained regions. Afterwards, the energy of the conformation is evaluated. First, the protein's structure in internal coordinates, is transformed into Cartesian coordinates. Then, the van der Waals and electrostatic energies are calculated using the TINKER routines and the hydrophobic formulation (Equation 3).

In order to reduce the size of the conformational space, the backbone torsion angles are constrained in regions derived from the CADB-2.0 [23] database, witch contains most torsion angles to each residue. The side-chain torsion angles are constrained in regions derived from the backbone-independent rotamer library of Tuffery [24]. Side-chain constraint regions are of the form: $[m - \sigma, m + \sigma]$, where m and σ are, respectively, the mean and the standard deviations for each side-chain torsion angle computed from the rotamer library. Under these constraints, the conformation is still highly flexible and the structure can take various shapes that are vastly different from the native shape.

At this point, we have the main loop of the algorithm. From current solutions, new solutions are obtained using genetic operators. We proposed three kinds of recombination operators: the first operator is the BLX-α operator especially developed for floating point representation [25]; the second operator uses the uniform crossover and the last is a two-point crossover. Each operator has its own application probability: 0.3 for first operator; 0.2 for second operator and 0.5 for third operator.

Three kinds of mutation operators were proposed. When the first mutation operator acts on a peptide chain, all the values of the backbone and side-chain torsion angles of a residue are chosen at random and then re-selected from their corresponding constrained regions. We made a copy of the parents individuals and modified this copied population M_1 times [26], each time by applying the operator to a randomly selected individual from this population. An individual can be selected more than once; consequently so there may be changes in torsion angles in more than one residue in a chromosome.

The second and the third mutation operator perturb all the values of the backbone and side-chain torsion angles of a selected residue using a uniform distribution. The number of perturbed residues of each individual is M_2 [26]. Both operators have equal probability to be employed. The difference between second and third operators is in the uniform distribution. Second operator uses random numbers in $[0.1]$, generated from a uniform distribution. Third operator is similar to second operator, the random numbers are in $[0, 0.1]$. The maximal perturbation of each individual is the standard deviation values to each group of dihedral angles of the side chain and the difference between 180 and the value of the angle in the main chain. These operators were also applied to every offspring in the reproduction process.

For the purposes of preventing premature convergence and random walk, we made M_1 and M_2 decrease as the search proceeds, as shown in Equation 5 and 6

$$M_1 = 1 + P \cdot exp(-gn/N_{eff}) \qquad (5)$$

$$M_2 = 1 + \frac{N}{4} \cdot exp(-gn/N_{eff}) \qquad (6)$$

where P is the population size, N is a number of residues in the protein, gn is the generation, and N_{eff} is a constant (set to 150).

V. Results

This section reports the results obtained using ProtPred. This algorithm was applied to four protein sequences from the Protein Data Bank[3] (PDB): 1A11, 1ALE, 1DEP and 1DU1. These protein are small α-helices which are complex to modeling with a pure *Ab initio* (see Section III) approach as ProtPred. In the sum of the energies function, that compose the EA fitness, was used a weight for each function. The population size is 500 chromosomes, the maximum number of generations is 100.

The cost function have dielectric constant equal to 4.0. In order to compute van der Waals and electrostatic energies, a maximum and a minimum distance (d) between atom i and atom j were defined. Only the atoms-interactions in this interval ($d_{min} \leq d \leq d_{max}$) are calculated. These bounds are specified in Table I.

TABLE I
PARAMETERS USED IN THE EA PRESENT IN PROTPRED

Parameter	Value
Population Size	500
Generations	100
Runs	1
Dielectric Constant	4
d_{max} electrostatic (Å)	13
d_{min} electrostatic	$1 * (radius_A + radius_B)^4$
d_{max} van der Waals (Å)	8
d_{min} van der Waals	$1.25 * (radius_A + radius_B)^5$
d_{max} hydrophobic (Å)	8.5
d_{min} hydrophobic (Å)	4.5
First recombination operator probability	0.3
Second recombination operator probability	0.2
Third recombination operator probability	0.5
Second mutation operator probability	0.5
Third mutation operator probability	0.5

In order to compute the hydrophobic interaction, a maximum distance between C_αs of different residues was defined to considered two residues as nearest neighbor. Moreover, a minimum distance between C_αs of different residues was also defined to determine if two residues are at the same position. Table I contains all parameters used in ProtPred.

Table II contains the weight sets used in ProtPred. The tests with ProtPred all proteins were performed for all weight sets presents in Table II.

Table III presents the weight sets used for each protein, the best fitness obtained, and the distance matrix error (DME) for each protein.

Membrane spanning segment of the acetylcholine receptor (1A11): is a single-helix peptide of 25 residues [27]. Figure 1 shows the 1A11 native structure and Figures 2, 3, 4, 5, 6 and 7 display the 1A11 predictions by our approach. Figure 2 shows the existence of two α-helix turns and a tendency to form other turns. The best predicted structure is

[3] www.rcsb.org/pdb

TABLE II
WEIGHT SETS USED BY PROTPRED.

Identifier	van der Waals	Eletrostatic	Hydrophobic
W1	1	1	2
W2	1	0.5	2
W3	1	0.5	1.75
W4	1	0.25	2
W5	1	0.25	1.5
W6	1	0.25	1.25
W7	1	0.25	1
W8	1	0	1.25
W9	1	0	1
W10	1	0	2
W11	1	0	1.5
W12	1	1	0

TABLE III
WEIGHT SET, MINIMUM FITNESS VALUE AND DME OF THE RELEVANT RESULTS.

Protein (PDB Id)	Weight set	Fitness	DME(Å)
$1A11$	W1	-262.9420	7.15
$1A11$	W2	-501.8676	1.05
$1A11$	W3	-481.7107	6.22
$1A11$	W4	-292.3069	7.89
$1A11$	W5	-258.5936	6.91
$1A11$	W12	-923.5357	4.74
$1ALE$	W1	-227.3685	4.54
$1ALE$	W6	-243.0533	4.11
$1ALE$	W7	-251.5571	5.08
$1ALE$	W8	-54.8493	4.78
$1ALE$	W9	-51.5810	4.96
$1ALE$	W12	-1455.7209	6.35
$1DEP$	W1	-237.1484	2.48
$1DEP$	W10	-62.4728	3.12
$1DEP$	W11	-61.3393	1.33
$1DEP$	W12	-1386.0300	5.00
$1DU1$	W1	-314.9491	3.77
$1DU1$	W12	-1865.2844	9.04

shown in Figure 3, where six α-helix turns can be seen like in the native structure.

Fig. 1. 1A11 Native Structure.

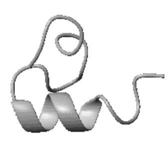

Fig. 2. 1A11 Predict Structure with W1 set.

These Figures show that the reduction of weights of electrostatic and hydrophobic energies in relation to van der Waals energy produces better predictions than ProtPred when uses W1 2. However, the best results is for W2, i. e., only the eletrostatic energy weigth was decreared. Thus, it seems that electrostatic energy can guide the search for inadequate local minima if it is relatively overweighted. On the other hand, its effect can not neglected in order to obtain adequate

Fig. 3. 1A11 Predict Structure using W2 set.

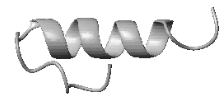

Fig. 4. 1A11 Predict Structure using W3 set.

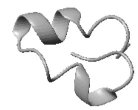

Fig. 5. 1A11 Predict Structure using W4 set.

Fig. 6. 1A11 Predict Structure using W5 set.

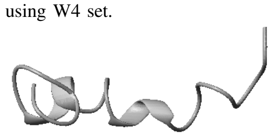

Fig. 7. 1A11 Predict Structure using W12 set.

prediction.

Apolipoprotein (1ALE): is a single-helix peptide composed of 18 residues from human synthetic fragments of apolipoprotein [28]. Figure 8 presents the 1ALE native structure and Figures 9 10, 11, 12, 13 and 14 show the predict 1ALE structures. The predictions have α-helix turns at the same position of the native structure.

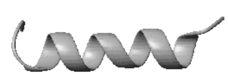

Fig. 8. 1ALE Native Structure.

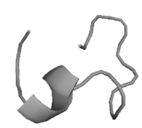

Fig. 9. 1ALE Predict Structure using W1 set.

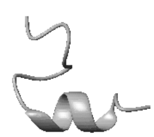

Fig. 10. 1ALE Predict Structure using W6 set.

Fig. 11. 1ALE Predict Structure using W7 set.

As in the results presented for 1A11 protein, reductions in electrostatic and hydrophobic weights (Figures 10, 11, 12 13 and 14) may result in better solutions, in the sense that more α-helix turns were achieved. However, for the set

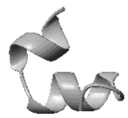

Fig. 12. 1ALE Predict Structure using W8 set.

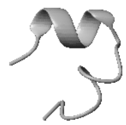

Fig. 13. 1ALE Predict Structure using W9 set.

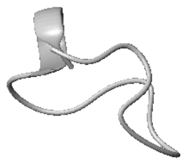

Fig. 14. 1ALE Predict Structure using W12 set.

of weights investigated, the ProtPred does not obtained an adequate prediction as was produced for 1A11.

Membrane protein (1DEP): is a 15-residue peptide with one α-helix [29]. 1DEP is synthetic peptide derived from the third intracellular loop of the beta-adrenoceptor. Figures 15, 16, 17, 18 and 19 present the native and predicted 1DEP structures, respectively. The predictions shown in Figure 16 has an α-helix folding with two turns and a tendency to form other turns. The predicted structure from Figure 18 is very similar to the native structure of 1DEP.

Fig. 15. 1DEP Native Structure.

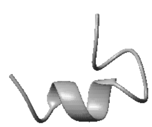

Fig. 16. 1DEP Predict Structure with W1 set.

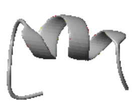

Fig. 17. 1DEP Predict Structure with W10 set.

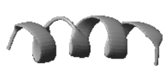

Fig. 18. 1DEP Predict Structure with W11 set.

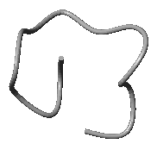

Fig. 19. 1DEP Predict Structure with W12 set.

In Figures 17 and 18 the electrostatic weight is 0, producing the best found structures in the sense that more α-helix turns were achieved. For 1DEP protein, as in 1ALE protein, the eletroctatic energy requires relatively low weight in order to ProtPred reach adequate predictions.

Peptide fragment of the THR671-LEU690 of the rabbit skeletal dihydorpyridine receptor (1DU1): is a single-helix peptide of 20 residues [30]. It is a helix from their N-terminus to lysine, however, is unstructured from lysine to the C terminus. Figure 20 presents the 1DU1 native structure and Figures 21 and 22 show the obtained structure, Figure 21 is very similar to the folding of the α-helix presented in the native structure. The other weight sets do not present better results to 1DU1 protein.

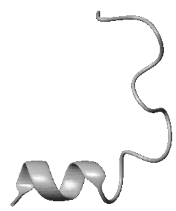

Fig. 20. 1DU1 Native Structure.

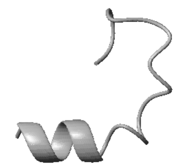

Fig. 21. 1DU1 Predict Structure using W1 set.

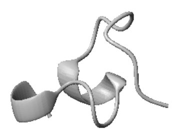

Fig. 22. 1DU1 Predict Structure using W12 set.

Analyzing all the tests perform, we can observe that different weight sets can produce better results for each different protein. Moreover, the values obtained of each energy function in the best solution of the population indicates that, when the electrostatic weight was reduced, the van der Waals energy increases its influence on the search process, enabling the better predictions. Table IV presents each energy value multiplied by its weight, for 1A11 protein, in order to verify this effect. When the hydrophobic interaction was not considered the resutls were worst than when it is considered in the fitness function (see Figures 7, 14, 19 and 22).

A. *Comparisons with other Ab initio approaches*

Approaches available in the literature for PSP problem [8] [19] have in general better results than ProtPred for the same proteins evaluated in this paper. Our proposal is to develop a pure *Ab initio* algorithm that does not use any heuristics in the prediction process, in contrast with *Ab initio* approaches like Rosetta [19] and the algorithms presented in [8]. Such approaches present relevant results to the PSP, using heuristics and information from the protein structure

TABLE IV

WEIGHT SET, VAN DER WAALS AND ELECTROSTATIC ENERGIES, AND HYDROPHOBIC INTERACTION FOR 1A11 PROTEIN.

Weight set	van der Waals	Electrostatic	Hydrophobic
W1	178.6673	-329.6093	-112.0000
W2	30.8524	-444.7200	-88.0000
W3	48.1271	-444.0878	-85.7500
W4	1.9659	-190.2729	-104.0000
W5	-8.7132	-173.3803	-76.5000

databases. Thus, since a priori knowledge is used, they can be characterized as *semi Ab initio* algorithms.

Another innovation of ProtPred is the use only of van der Waals and electrostatic energies and the addition of hydrophobic interaction calculated from a HP model with a full-atom model. The other approaches for PSP with full-atom model do not use hydrophobic energy to evaluate their solutions, they employ only potential energies [3].

It is important to highlight that, a pure *Ab initio* approach can explore any region of the search space to find an adequate structure conformation even though there is no similar protein structure previously known.

VI. CONCLUSIONS

This paper has presented ProtPred algorithm for PSP. ProtPred evaluates the obtained solutions using: van der Waals energy, electrostatic energy and hydrophobic interactions.

The novelties in ProtPred is development of a pure *Ab initio* approach to PSP and the usage of different evaluation of solutions that combine van der Waals and electrostatic energies with hydrophobic interactions. The most of evolutionary *Ab initio* full-atom approaches uses only potential energies to evaluate the solutions obtained. Moreover, the *Ab initio* approaches found in the literature are not pure since heuristics are employed to guide their search algorithms.

The results found indicated that ProtPred can predict small proteins structures with relative efficiency. Thus ProtPred is a consistent approach to PSP problem that can predict protein tertiary structures without previously heuristics. As future work, we suggest a more investigation of potential energies interactions and their effects in the protein folding.

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